

MTH 206 INTRODUCTION TO ODE

Midsem 1

Max Marks: 15

Time: 1 hour

- (1) Let f_n be a solution of the boundary value problem

$$y'' = -n^2(x^2 + 1)y, n \geq 0, x \in [0, 1]$$

and $y(0) = y(1)$; $y'(0) = y'(1)$. Show that

$$\int_0^1 f_n(x)f_m(x)(x^2 + 1)dx = 0$$

for $n \neq m$.

[5]

- (2) Let ϕ_1, ϕ_2 be two differentiable functions on an interval I such that $W(\phi_1, \phi_2) \equiv 0$ on I . Let ϕ_1 be non zero on I . Show that ϕ_1, ϕ_2 are linearly dependent on I .

[5]

- (3) Let $L(y) = y'' + a_1y' + a_2y$ be second order, linear homogeneous ODE. Let ϕ satisfy the $L(\phi) = 0$ on $[0, \infty)$. Find the conditions on a_1 and a_2 such that $|\phi(x)| \rightarrow 0$ as $x \rightarrow \infty$.

[5]